

CM/LM Research Consolidation

Canonical Theorem Bank and Evidence Boundaries

Research consolidation for Brian Theory

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Abstract

This bank separates the existing Boolean operator calculus from a chosen rotation algebra, explicit modal resource contracts, exact query models and symbolic support interfaces. It consolidates proofs rather than treating historical computation as authority. The main candidate contributions are a finite-chain-ring contextuality classification, its leading-layer codebook synthesis, and exact characteristic-two query bounds. Priority is not certified. Background imports, finite-only observations and missing empirical evidence are identified explicitly. Companion registries carry source locations, test scope and one-primary-paper ownership for 71 claims.

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1 Definitions and publication conventions

All sums of Boolean coefficients are XOR; products of Boolean scalars are AND. Let $\mathcal{B}$ be a Boolean algebra of formulas modulo logical equivalence, regarded as a Boolean ring. Its elements satisfy $b^2 = b$. The variables X, Y used in the fixed LM family are independent formal generators. Paper B uses the true-first row and column order $(1, 0)$, hence row-major truth order $(11, 10, 01, 00)$. The coefficient rotation instead uses the cyclic order $(11, 10, 00, 01)$. Converting between these orders is part of a map, not an implicit identification.

Let $Re_j = e_{j+1 \pmod{4}}$, $u = I + R$, and $\mathcal{A} = \mathbb{F}_2[R] \cong \mathbb{F}_2[u]/(u^4)$. Elements of $\mathcal{A}$ have four binary coefficients; matrices over $\mathcal{A}$ are not original 2×2 correspondence matrices. The faithful binary regular representation of $M \in \text{Mat}_d(\mathcal{A})$ is denoted $\lambda_d(M) \in \text{Mat}_{4d}(\mathbb{F}_2)$.

More generally, K is a finite commutative chain ring with maximal ideal (π) , nilpotency length s and residue field $k = \mathbb{F}_q$. Set $\pi^s = 0$. For a nonzero matrix with Smith exponents $0 \leq a_1 \leq \dots \leq a_\ell \leq s$, let

$$r = \#\{i : a_i < s\}, \quad h = \#\{i : a_i = a_1\}.$$

Here r means the number of nonzero Smith factors, not an unspecified ring rank; h is the least-valuation multiplicity. A row is primitive when one coordinate is a unit. Primitive rows modulo unit multiplication are projective rows. A complete local setting is a projective basis represented by rows of an invertible matrix. Different bases are different settings; no extra context-independent identification of repeated rays is assumed.

For a bipartite resource M , the supported events are

$$\mathcal{E}_M(B, C) = \{(i, j) : b_i M c_j^T \neq 0\}.$$

A global assignment selects one outcome in every local setting and is supported in every joint setting. The support is *local* when every supported event extends to a global assignment, *logical but not strong* when some event does not extend but a global assignment exists, and *strong* when none exists. The zero resource, which has empty support, is excluded from the classification of physical-looking nonzero resources.

A status of “proved” addresses validity under stated assumptions, not priority. A finite record below is never an unbounded theorem. Source abbreviations S-B, S-AUD, S-ADV, S-CAP and S-LMB resolve in `provenance/SOURCE_GUIDE.md`. Candidate novelty means only that the bounded targeted search did not settle priority. Stable claim IDs resolve in `claims/claim_registry.json`.

2 Operational contracts

F: field modal model. States are nonzero vectors of $\mathbb{F}_2^d$, composite spaces use $\otimes_{\mathbb{F}_2}$, and the allowed reversible group is stated explicitly. An unrestricted complete readout is a dual basis; an outcome is possible exactly when its coefficient is nonzero. No real probabilities or positive inner product are supplied.

S: shared-ring model. Local carriers are K^d , composites use $\otimes_K$, and complete local measurements are primitive projective bases. Static resource tensors are nonzero. This is not yet a closed all-nonzero preparation theory: nonzero tensor factors can annihilate. A primitive-only preparation restriction repairs that problem but is not closed under the specified conditioning rule.

L: literal registers. A logical-plus-phase carrier is $\mathbb{F}_2^2 \otimes_{\mathbb{F}_2} \mathbb{F}_2^4 = \mathbb{F}_2^8$. Its n independent copies have dimension 8^n , whereas the binary dimension of $(\mathcal{A}^2)^{\otimes \mathcal{A}^n}$ is $4 \cdot 2^n$. Literal restricted coarse readouts and unrestricted $\text{GL}_8(\mathbb{F}_2)$ readouts are different contracts. The regular representation is a faithful matrix-algebra map, not a monoidal equivalence of these state spaces.

DC: exact fixed-orbit codebooks. For $M \in \text{Mat}_d(K) \setminus \{0\}$, $d \geq 2$, Alice encodes by $U \in \text{GL}_d(K)$, giving UM . Bob has one arbitrary complete invertible K -linear basis readout on the d^2 joint coefficients. Every possible outcome must identify the message. Discarding the fixed resource and preparing a fresh carrier is outside this orbit count.

TP: exact universal rank-one teleportation. For every nonzero input, including an arbitrary untouched reference system, all possible branches of a complete joint rank-one analyzer must be corrected exactly. In a fixed vectorization convention the transfer is $T_m = M^T E_m^T$. A complete basis of flattened E_m and invertibility of every reshape E_m are separate requirements. Allowed corrections must contain the necessary inverses. No unrecorded failures or input-dependent environment may be ignored.

QXOR and its lower-bound extension. The standard oracle is $U_f|x, y\rangle = |x, y + f(x)\rangle$ over a characteristic-two field. Arbitrary finite ancillas and oracle-independent linear processing are allowed. Complete adaptive instruments can be recorded in a direct sum; the collected readout is injective and every possible outcome must be correct. Classical inspection of a state description or its complete support is not free readout. The lower bounds extend to $O_f = \sum_x |x\rangle\langle x| \otimes G_{f(x)}$ for fixed invertible G_0, G_1 . Matching upper bounds are asserted for QXOR, not arbitrary G_0, G_1 : the choice $G_0 = G_1$ makes all functions indistinguishable.

PROB: additional real completion problem. A real probability completion is nonnegative, normalized and no-signalling. It is faithful only if every supported event has positive probability. A weak completion can delete possible events by assigning them zero. Neither type is inferred automatically from modal support.

3 Imported Paper B foundations

Canonical IDs: CM-B-001–007.

Proposition 3.1 (Imported CM selection and aligned fusion). *Let $s(x) = (x, 1 + x)^T$ and let C_f have entries $f(r, t)$ in true-first order. Then $s(x)^T C_f s(y) = f(x, y)$. If two occurrences have the same declared ordered operand frame, for every binary Boolean connective g ,*

$$g(f_1(x, y), f_2(x, y)) = s(x)^T (g(C_{f_1}, C_{f_2})) s(y),$$

where g acts entrywise. Operand exchange transposes the CM; input negations permute the corresponding axis.

Provenance and status. S-B sections 2–4 and appendix A.1–A.4. Imported, not re-proved as later novelty. One-hot selection is the reason arbitrary g is valid; no distributive law of g over XOR is needed. Structural frame admission, retabulation and fallback remain different compiler paths.

Proposition 3.2 (Imported symbolic pairing). *The formula-valued LM is*

$$M_f(X, Y) = \begin{pmatrix} f(X, Y) & f(X, \neg Y) \\ f(\neg X, Y) & f(\neg X, \neg Y) \end{pmatrix}.$$

For any formulas P, Q , its logical pairing is

$$s(P)^T M_f(X, Y) s(Q) \equiv f(P \leftrightarrow X, Y \leftrightarrow Q).$$

Boolean valuations commute with this pairing. Positive valuation $X = Y = 1$ recovers C_f .

Provenance and status. S-B section 8, Proposition 2 and Theorem 3; appendix A.5. These are Boolean formulas, not an already selected modal measurement theory. A four-bit token and an explicit 2^n -entry truth relation have different cost contracts.

4 Rotation lift: classical algebra and its semantic boundary

Canonical IDs: ROT-001, ROT-002, ROT-007.

Theorem 4.1 (Cyclic commutant and chain ring). *For the regular N -cycle on $\mathbb{F}_2^N$, the centralizer is $\mathbb{F}_2[R] \cong \mathbb{F}_2[x]/(x^N - 1)$. If $N = 2^n$ and $u = I + R$, this is $\mathbb{F}_2[u]/(u^N)$. Multiplication by $u^a v$, for a unit v , has binary rank $N - a$, $0 \leq a \leq N$. A coefficient polynomial is a unit iff its coefficient parity is one. Consequently*

$$\text{End}_{C_N}((\mathbb{F}_2^N)^m) = \text{Mat}_m(\mathbb{F}_2[R]), \quad \text{Aut}_{C_N}((\mathbb{F}_2^N)^m) = \text{GL}_m(\mathbb{F}_2[R]).$$

Proof. A commuting operator is determined by its first column v : its j th column must be $R^j v$. The N powers of R are independent by application to e_0 , so the only defining relation is $R^N = I$. Frobenius gives $(1 + x)^N = 1 + x^N$. In the u basis, multiplication by u^a shifts $N - a$ basis vectors to independent nonzero vectors and kills the remaining a . Invertibility is equivalent to a nonzero constant u coefficient, namely the original polynomial evaluated at one. The block commutant statement applies the same condition to every block. $\square$

Provenance and status. S-ADV D1 and D3. Standard group-algebra/commutant facts, not new CM algebra. The maximal equivariant gate group is derived; support, tensors and preparation rules are not.

Canonical IDs: ROT-003, ROT-005.

Corollary 4.2 (Full-degree criterion and distinct filtrations). *For an n -variable Boolean function f , $n \geq 1$, its cyclic coefficient lift of length 2^n is invertible iff $\bigoplus_x f(x) = 1$, equivalently the ANF coefficient of $x_1 \cdots x_n$ is one, equivalently $\deg f = n$. This does not identify the whole cyclic valuation filtration with ANF degree.*

Proof. Summing a proper-support monomial over all assignments gives zero in $\mathbb{F}_2$; the full monomial contributes one. Apply the unit criterion. For binary-indexed coefficients a_j ,

$$\sum_j a_j (1 + u)^j = \sum_k b_k u^k, \quad b_k = \bigoplus_{j \supseteq k} a_j.$$

The cyclic valuation uses the least *numeric* k with $b_k \neq 0$. By contrast the elementary-abelian group algebra $\mathbb{F}_2[u_1, \dots, u_n]/(u_i^2)$ uses total degree, the Hamming weight of k . For an ANF monomial with support S , its truth coefficient polynomial there is $\prod_{i \in S} (1 + u_i) \prod_{i \notin S} u_i$; the least degree is $n - |S|$. Distinct highest-degree ANF monomials give distinct least-degree terms. Thus its radical order is $n - \deg f$. For a four-cycle the affine coefficient vectors $(1, 1, 0, 0)$ and $(1, 0, 1, 0)$ have cyclic orders one and two, respectively. $\square$

Provenance and status. S-ADV D1–D2. The Reed–Muller/radical identification is classical Berman–Charpin theory, not a new finding [14]. Cyclic order also encodes the standard periodic linear-complexity invariant $N - \deg \gcd(a(x), x^N - 1)$. Original Games–Chan priority details remain an inherited bibliographic lead, not a newly read source.

Canonical IDs: ROT-004, ROT-006, ROT-009.

Proposition 4.3 (Reversal, complement and XOR interferometer). *For $Ke_j = e_{-j}$, $KRK = R^{-1}$ and $K\Lambda(c)K = \Lambda(c)^T$. In the specified CM cyclic order this reflection exchanges the two Boolean arguments. Complementing a length- $N = 2^n$ truth vector adds u^{N-1} . With*

$$W = \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}, \quad D_k = \text{diag}(I, R^k),$$

we have $W^2 = I$ and $WD_kW(v, 0)^T = (v, v + R^kv)^T$.

Proof. Reflection sends each power R^j to $R^{-j} = (R^j)^T$. Complement adds the sum of all powers; at length 2^n this is $(1 + R)^{N-1}$. The displayed interferometer identity follows by three block multiplications. $\square$

Scope. Complement is translation, not multiplication by the socle. Reversal is not generally operator inversion, though they coincide on the eight units of $\mathcal{A}$ at length four. R is unipotent, not multiplication by the complex number i . Its eigenvalues in field extensions are all one; a three-dimensional unipotent block can already have order four. The ordinary identity CM lifts to $I + R^2 = u^2$, whose square is zero, so this lift cannot preserve the original CM matrix product or entrywise AND.

Provenance and status. S-ADV D1–D3; S-CAP F1. Standard identities used as typed motivation.

5 Complete-basis contextuality over chain rings

Canonical IDs: MOD-003.

Lemma 5.1 (Residue-hyperplane criterion). *For the complete primitive projective basis support model over a finite commutative local ring, a global assignment exists iff there are hyperplanes $H_A \subset k^m$, $H_B \subset k^n$ such that $rMt^T \neq 0$ for every primitive lift r, t of vectors outside H_A, H_B , respectively.*

Proof. Given a global assignment, let U_A be the rays selected at least once among Alice’s settings. The complement of U_A contains no basis: such a setting would have to select a ray in its own

complement. If the reductions of the complement spanned k^m , choose a residue basis from them; its lifts would be an invertible local-ring basis, a contradiction. Put their span in a hyperplane H_A . Every lift of a residue ray outside H_A belongs to U_A . Do the same for Bob. Every cross pair from U_A, U_B is selected in some pair of settings and must have nonzero contraction. Conversely, each invertible basis has a row reducing outside any fixed hyperplane. Selecting one such row in each setting on each side gives a compatible assignment by hypothesis. $\square$

Provenance and status. S-ADV D4 Lemma 8. Candidate formulation; priority unresolved. This lemma uses locality but not the chain condition. The independent dual-number check uses explicit support and 2-SAT rather than this criterion.

Canonical IDs: MOD-005.

Lemma 5.2 (Valuation-uniform Hardy event). *For $D = \text{diag}(p, q_0)$ with $p = \pi^a$, $q_0 = \pi^b \neq 0$, $a \leq b$, put $c = \pi^{b-a}$, so $pc = q_0$. The reversible basis matrices*

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad C_c = \begin{pmatrix} 1 & 0 \\ -c & 1 \end{pmatrix}$$

give, for Alice settings F, C_c and Bob settings F, X ,

$$\begin{aligned} FDF^T &= \begin{pmatrix} q_0 & 0 \\ 0 & p \end{pmatrix}, & FDX^T &= \begin{pmatrix} 0 & q_0 \\ p & p \end{pmatrix}, \\ C_cDX^T &= \begin{pmatrix} p & p \\ -q_0 & 0 \end{pmatrix}, & C_cDF^T &= \begin{pmatrix} 0 & p \\ q_0 & -q_0 \end{pmatrix}. \end{aligned}$$

The supported event ($F_A = 0, F_B = 0$) has no global extension.

Proof. The determinants are units and direct multiplication gives the matrices. The seed forces $X_B = 1$, then $C_A = 0$, then $F_B = 1$, contradicting the seed. The support strings are 1001, 0111, 1110, 0111. $\square$

Provenance and status. S-ADV D5. The Hardy implication pattern is known; the uniform valuation-dependent implementation is part of the candidate chain-ring classification. The minus sign is essential outside characteristic two.

Canonical IDs: MOD-004.

Theorem 5.3 (Smith contextuality trichotomy). *For a nonzero $m \times n$ matrix over K , the complete-basis support type is*

$$\left. \begin{array}{l} r = 1 \\ r \geq 2, h = 1 \\ h \geq 2 \end{array} \right| \begin{array}{l} \text{local} \\ \text{logical but not strong} \\ \text{strong.} \end{array}$$

Proof. Invertible left and right transformations relabel the complete basis families, so take Smith form. If $r = 1$, write $M = \text{diag}(p, 0, \dots)$ and retain the outcomes of any supported seed event. In all other settings choose a row whose first coordinate is a unit. New-new pairs are nonzero multiples of p . A seed-new pair is also nonzero: if $pr_1t_1 \neq 0$, then neither pr_1 nor pt_1 is zero, and multiplication by a unit preserves nonzeroness. Thus every supported seed extends.

If $h = 1$, in every basis select a first-coordinate-unit row. Write $M = \pi^{a_1} \text{diag}(1, \pi^{a_2 - a_1}, \dots)$. The contraction in parentheses is a unit plus radical terms, so the selected cross pair is nonzero. Hence a global assignment exists.

If $h \geq 2$, suppose the hyperplanes in the preceding criterion existed and write $M = \pi^a M_0$ with $\text{rank}_k \overline{M}_0 = h$. Let $H_B = \ker \beta$. The vectors outside H_A span k^m . Therefore some x outside H_A has $w = x\overline{M}_0$ not in the one-dimensional span of β . There is $y \in \ker w$ outside $\ker \beta$. Lift x, y to r, t . The row rM_0 has a unit coordinate; $rM_0 t^T$ lies in the radical. Adjust the corresponding coordinate of t by a radical element to cancel that dot product exactly. The residues remain x, y , but $rMt^T = 0$, contradicting the hyperplane criterion. Thus no global assignment exists.

Finally, if $r \geq 2$, embed the preceding two-coordinate Hardy construction in the first two nonzero Smith coordinates and extend the settings with standard rows. The seed and each forcing step remain in those coordinates; extra outcomes have zero cross entries with the forced rows and cannot evade the contradiction. Thus some event never extends. The three cases follow. $\square$

Provenance and status. S-ADV D4 Theorem 9; imported and independently checked in S-CAP R1. Candidate novelty, not certified priority. The proof applies to mixed-characteristic chain rings as well as polynomial quotients; the symbolic $\mathbb{F}_2$ enrichment later has a narrower scalar scope.

Canonical IDs: MOD-006.

Corollary 5.4 (Basis counts). *The numbers of primitive projective rows and unordered projective bases in K^m are*

$$q^{(s-1)(m-1)} \frac{q^m - 1}{q - 1}, \quad \frac{q^{m^2(s-1)} |\text{GL}_m(q)|}{|K^\times|^m m!},$$

respectively. For $\mathcal{A}^2$ they are 24 and 192; $|\text{GL}_2(\mathcal{A})| = 24576$.

Proof. There are $q^{sm} - q^{(s-1)m}$ primitive vectors. The unit action is free on them and $|K^\times| = q^{s-1}(q - 1)$. An invertible residue matrix has $q^{m^2(s-1)}$ lifts, each invertible. Quotient ordered invertible rows by their independent unit scales and by the free row-permutation action. $\square$

Provenance and status. Standard finite local-ring counting; S-ADV D4.

6 Tensor choice, resources and exact protocols

Canonical IDs: MOD-001, MOD-002, MOD-009, MOD-010.

Proposition 6.1 (Tensor and orbit boundaries). *In $\mathcal{A} \otimes_{\mathcal{A}} \mathcal{A}$, $u^3 \otimes u = 0$; in $\mathcal{A} \otimes_{\mathbb{F}_2} \mathcal{A}$ it is nonzero. Primitive preparations remain primitive under a balanced tensor, but conditioning the primitive resource $\text{diag}(1, u)$ by the second standard effect gives the nonprimitive nonzero vector $(0, u)$. For Smith exponents (a_i) ,*

$$\text{rank}_{\mathbb{F}_2} \lambda_d(M) = \sum_i \max(4 - a_i, 0).$$

Unrestricted local binary matrix equivalence is classified only by this rank.

Proof. Balanced tensor multiplication identifies the first scalar tensor with $u^4 = 0$. A nonzero simple tensor over a field is nonzero, as witnessed by two nonvanishing dual functionals. A primitive vector has a unit coordinate; the tensor of two such coordinates is a unit. The stated conditional vector is obtained by selecting the second matrix row. Regular expansion of Smith form is block diagonal; each u^{a_i} block has rank $4 - a_i$, and invertible Smith changes remain invertible after expansion. Field matrix equivalence by independent invertible left/right changes has ordinary rank normal form. $\square$

Scope. $D(0, 2)$ and $D(1, 1)$ both have binary rank six but different restricted contextuality. The latter is nonprimitive. Primitive dimension-three examples $\text{diag}(1, 1, u^3)$ and $\text{diag}(1, u, u^2)$ both have binary rank nine but different leading multiplicities. The particular audited restricted shared/literal *bipartite support tables* are nevertheless isomorphic: literal Choi contraction transposes Bob's blocks, and $R \mapsto R^{-1}$ permutes the admitted basis family. This does not make the tensor theories equivalent.

Provenance and status. S-ADV D6; S-AUD section 7.4; S-CAP F2. Standard algebra plus a model-specific restriction comparison.

Canonical IDs: ALG-001.

Lemma 6.2 (Exact linear discrimination). *Let S_i be nonzero state sets in a vector space over a field and $U_i = \text{span}(S_i)$. A complete injective linear readout can distinguish the labels with every possible outcome correct iff $\sum_i U_i$ is a direct sum.*

Proof. The coordinates assigned to different labels must be disjoint. The image of U_i therefore lies in its own coordinate subspace. Injectivity of the collected readout forbids a nontrivial dependence between these subspaces. Conversely, concatenate bases of the independent U_i and extend to a basis of the entire space; the resulting dual-basis readout distinguishes the labels. $\square$

Provenance and status. S-CAP Q1. The single-state independence criterion is modal/linear-algebra background; see also [10]. Its uncertainty-set and recorded-branch form is a useful supporting formulation, not the P02 headline.

Canonical IDs: RES-001, RES-002.

Theorem 6.3 (Leading-layer exact codebook). *Under contract DC, the largest exactly distinguishable invertible-orbit codebook for $M \in \text{Mat}_d(K) \setminus \{0\}$, $d \geq 2$, has dh messages. Consequently advantage over the d -message baseline is equivalent to strong contextuality under the same complete chain-ring basis contract.*

Proof. Write $M = \pi^a N$ with $\text{rank}_K \overline{N} = h$. The reduction of a divided codeword UN lies in $\{X\overline{N} : X \in \text{Mat}_d(k)\}$, a vector space of dimension dh . Every encoded resource retains a nonzero leading layer. After any invertible joint decoder, exactly distinguishable codewords have disjoint nonzero supports; their leading residue vectors are therefore independent. This proves the upper bound dh .

Invertible $d \times d$ matrices over a field span the full matrix algebra. Off-diagonal units are differences of $I + E_{ij}$ and I . For a diagonal unit, choose a permutation P sending some $j \neq i$ to i ; the difference of $P(I + E_{ji})$ and P is E_{ii} . Both matrices are invertible. Thus one can choose dh invertible residue encoders for which $\overline{U}_m \overline{N}$ are independent. Lift them to invertible matrices over K . The columns $\text{vec}(U_m N)$ are independent modulo the radical, so extend to a basis modulo the radical and lift the

remaining columns. The full lifted matrix is invertible. Its inverse decodes the actual codeword $\pi^a \text{vec}(U_m N)$ to $\pi^a e_m$, with exactly one possible outcome. This attains dh . The contextuality equivalence follows from the trichotomy, since $dh > d$ iff $h \geq 2$. $\square$

Provenance and status. S-CAP R2 and S-ADV D4. Candidate theorem/synthesis pending priority clearance. All 14 nonzero $\mathcal{A}$ Smith representatives have explicit independently checked decoder matrices. This counts messages in a fixed orbit, not entropy.

Canonical IDs: RES-003, RES-004, RES-007.

Theorem 6.4 (Universal exact teleportation and its separation). *Under TP, every nonzero universally correct branch must have invertible transfer $T_m = M^T E_m^T$. Hence M must be invertible. Conversely, if M is invertible and a complete basis of invertible reshaped effects is admitted, correction by T_m^{-1} is exact, including reference correlations. For $0 < a < s$, $\pi^a I_d$ has d^2 DC messages and strong contextuality, but cannot satisfy this teleportation contract.*

Proof. If a nonzero branch has nonzero kernel vector k , choose x with $T_m x \neq 0$. Then x and $x + k$ are distinct nonzero inputs with the same possible branch output. No fixed correction can recover both exactly. Thus a nonzero universal branch is injective. On a finite equal-rank free module an injective square map is bijective and has a linear inverse. Since T_m factors through M^T , its invertibility implies that of M . Completeness and a nonzero resource forbid all branches being identically zero. In the sufficient case the identity $T_m^{-1} T_m = I$ proves the claim; tensoring it with a reference identity preserves correlations. Finally $\pi^a I_d$ has leading multiplicity d but annihilates nonzero vectors, so apply the codebook theorem and necessity. $\square$

Scope. Exact raw-vector recovery is the stated task, not an unspecified projective equivalence relation. A singular resource is not rescued by selecting one purported nonzero universally correct branch. The nilpotent separator is nonprimitive and depends on allowing that preparation.

Provenance and status. S-CAP R3; S-ADV D9. The general transfer identity is established modal background. Its chain-ring resource comparison is the P01 synthesis.

Proposition 6.5 (Explicit admitted analyzers in the supplied carriers). *The four matrices*

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

are invertible and their flattened rows form a basis. They give a complete four-outcome analyzer over $\mathcal{A}$. Their triple field tensor products give a complete 64-outcome analyzer for a literal eight-dimensional carrier.

Proof. Each determinant is one. If the coefficients of a linear relation are $\alpha, \beta, \gamma, \delta$, the four entries successively give $\delta = \beta$, $\gamma = \alpha$, $\beta = 0$, $\alpha = 0$. A tensor product of bases is a basis and a tensor product of invertible matrices is invertible. The correction follows from the preceding theorem. $\square$

Scope. Do not identify a four-outcome logical analyzer with a complete analyzer for the literal phase register. General-dimensional analyzer sufficiency can be developed using the invertible-matrix spanning lemma, but the canonical supplied protocol claim here retains its explicitly admitted carriers.

Canonical IDs: RES-005, RES-006, RES-008.

Proposition 6.6 (Field codebooks and literal extraction). *For a rank- r matrix in $\text{Mat}_d(F)$, $d \geq 2$, unrestricted invertible-orbit encoding and complete field-linear joint readout give exactly dr codewords. A rank- r resource admits local rectangular filters extracting I_t from k field-tensor copies iff $r^k \geq t$.*

Proof. The image of $X \mapsto XM$ has dimension dr . Invertible matrices span $\text{Mat}_d(F)$ by the preceding argument; select a basis from their images and apply the discrimination lemma. For extraction, reduce M to rank normal form. Its k th tensor has rank r^k ; select t nonzero coordinate directions when possible. Conversely rectangular multiplication cannot increase rank. $\square$

Scope. For a regular 2×2 $\mathcal{A}$ resource, restricting encoders to $\text{GL}_2(\mathcal{A})$ but allowing a fine unrestricted binary decoder gives $2r_{\text{bin}}$ codewords (S-CAP R4). This uses the binary span of $\text{Mat}_2(\mathcal{A})M$ and is an asymmetric restriction. It can lie below the ordinary eight-letter baseline; allowing discard/reprepare gives at least $\max(8, 2r_{\text{bin}})$. Extraction is an algebraic filter statement, not a distillation probability.

7 Setting and form restrictions

Canonical IDs: MOD-007.

Theorem 7.1 (Two complete binary settings are not strongly contextual). *For any number of parties, a nonzero $\mathbb{F}_2$ tensor with at most two complete binary basis settings per site has a global assignment. The same holds for two-dimensional local modules over any finite commutative local ring with residue field $\mathbb{F}_2$, using nonzero-amplitude support.*

Proof. Two distinct binary dual bases share a row a ; write the other rows as b and $a + b$. Expand the tensor in coordinates dual to each pair (a, b) . Choose an inclusion-minimal set S of b coordinates having nonzero coefficient. Outside S , select a in both settings. Inside S , select b in the first and $a + b$ in the second. Each joint choice evaluates to the coefficient for S plus coefficients of proper subsets of S . By minimality the latter are zero and the former is one. Every chosen event is therefore supported. Coincident settings can be duplicated and require no additional choice.

For a finite local ring with radical J , take a nonzero leading layer of the tensor in J^a/J^{a+1} . Apply an $\mathbb{F}_2$ -linear functional on that layer which makes the resulting coefficient tensor nonzero. Reduce the local bases modulo J and apply the field construction. Each selected contraction then has a nonzero leading-layer functional, so its original ring coefficient is nonzero. $\square$

Scope. Literal independent phase blocks can be organized in the local algebra $\mathbb{F}_2[u_1, \dots, u_n]/(u_i^4)$; the same leading-layer proof applies to the specified coarse measurements without identifying it with shared $\mathcal{A}$. This theorem permits two-setting Hardy logical contextuality and does not cover three settings, higher outcomes or arbitrary residue fields.

Provenance and status. S-CAP R1b. Candidate general theorem. The source algorithm checked 65,808 nonzero states for one through four mobits and 1,050,666 supported-context checks; the proof, not that enumeration, handles arbitrary party count.

Canonical IDs: RES-009, ROT-008.

Proposition 7.2 (Transpose-orthogonal span and a separate form obstruction). *For $d \geq 2$,*

$$\text{span}_{\mathbb{F}_2} O_d(\mathbb{F}_2) = \{M : M\mathbf{1} = c\mathbf{1}, \mathbf{1}^T M = c\mathbf{1}^T \text{ for some } c \in \mathbb{F}_2\},$$

of dimension $(d-1)^2 + 1$. Therefore the stated complete rank-one teleportation analyzer with transpose-orthogonal corrections cannot exist. Separately, the native W and $\text{diag}(I, R)$, $R \neq I$, preserve no common nondegenerate bilinear form on the same doubled space.

Proof. Every column of an orthogonal binary matrix has odd weight; therefore both its row and column sums are one. Linear combinations lie in the displayed space. Permutation matrices are orthogonal. Differences of permutations produce each four-corner rectangle involving a fixed last row and column; these span the $(d-1)^2$ -dimensional zero-row/column-sum space. Adjoining I adds the common-sum direction. For a fixed invertible teleportation resource, reshaped effects with orthogonal corrections lie in an invertible translate of this span, too small to span the d^2 -dimensional dual space.

For the second statement write a form matrix as $\begin{pmatrix} A & C \\ E & F \end{pmatrix}$. Invariance under W gives $F = 0, E = C$. Nondegeneracy forces C invertible, since $(0, x)$ is a null vector when $Cx = 0$. Invariance under $\text{diag}(I, R)$ gives $CR = C$, hence $R = I$, a contradiction. $\square$

Provenance and status. S-ADV D7–D8 and S-CAP E1–E2. The permutation-span ingredient is classical [13]; the dimension-eight span is 50, not merely bounded by 63. These are scoped model obstructions, not universal claims about all symplectic or finite-field theories. R itself preserves the alternating form R^2 ; different admissible gate families remain possible.

8 Characteristic-two exact query theory

Canonical IDs: ALG-002.

Theorem 8.1 (Deutsch: a lower bound and its properly typed upper bound). *With any fixed invertible pointwise G_0, G_1 over a characteristic-two field, no exact one-query linear modal procedure computes $f(0) + f(1)$. For the standard XOR oracle QXOR the exact query complexity is two.*

Proof. Write the prepared state by address sectors (a_0, a_1) , allowing all target and ancillary degrees of freedom. The four postquery states satisfy

$$v_{00} + v_{11} = v_{01} + v_{10} = ((G_0 + G_1)a_0, (G_0 + G_1)a_1) = w.$$

If $w \neq 0$, the constant and balanced spans intersect nontrivially, contrary to exact discrimination. If $w = 0$, both sector differences vanish and all four states are identical. Complete subsequent linear readout cannot remove either obstruction. Under QXOR, query both addresses on basis labels, retain the two answers and compute their XOR; this attains two. $\square$

Scope. The upper bound is not valid for arbitrary G_0, G_1 without an informative interface. If $G_0 = G_1$, every oracle is identical at every query count. This explicit qualification is a consolidation correction to the broad wording of S-CAP Q2.

Provenance and status. S-CAP Q2, read with the QXOR contract; ALG-002; correction C17. Candidate exact-query result after dedicated but nonexhaustive prior-art search.

Canonical IDs: ALG-003.

Theorem 8.2 (Deutsch–Jozsa one-query obstruction). *For $n \geq 2$ and the constant-or-balanced promise on $N = 2^n$ addresses, the same pointwise characteristic-two contract has no exact one-query solution. For QXOR the established bounds are $2 \leq Q_{\text{exact}} \leq N/2 + 1$.*

Proof. Partition the addresses into four equal blocks A, B, C, D . The balanced functions with supports $A \cup B$, $A \cup C$, $B \cup C$ have pointwise XOR zero. The pointwise oracle is affine in its control bit, so their three oracle operators sum to O_0 in characteristic two. The nonzero constant-zero postquery state is therefore in the balanced-state span, violating exact discrimination. The QXOR upper bound queries $N/2 + 1$ distinct addresses: agreement forces constancy, and any difference proves balance. The $n = 1$ case is Deutsch. $\square$

Scope. No claim is made that the classical upper bound is optimal for all multi-query modal procedures. One-query failure is stronger than failure of a particular Hadamard circuit, but is not an all-algorithm lower bound of $N/2 + 1$.

Provenance and status. S-CAP Q3. Candidate query result; source checks cover 56 operator coordinates, not every ancillary state. The proof supplies ancillary generality.

Canonical IDs: ALG-004.

Theorem 8.3 (Exact Bernstein–Vazirani lower bound). *For $f_s(x) = s \cdot x$, $s \in \mathbb{F}_2^n$, a q -query exact characteristic-two linear modal algorithm under the fixed pointwise oracle requires $q \geq n$. Under QXOR, n basis-label queries attain the bound.*

Proof. The oracle is affine in the secret: $O_s = O_0 + \sum_j s_j A_j$. After q queries, every state component is a multilinear polynomial of degree at most q in the bits of s (replace s_j^2 by s_j). Thus all 2^n final states lie in the span of at most

$$\sum_{j=0}^{\min(q,n)} \binom{n}{j}$$

coefficient vectors. For $q < n$ this is strictly less than 2^n , while exact separation of the 2^n singleton labels requires independent final states. Ancillas enlarge coefficient vectors, not their number. Recording all complete adaptive branches in a direct sum preserves both the degree bound and the injectivity requirement. QXOR queries at e_j directly return s_j . $\square$

Provenance and status. S-CAP Q4. Polynomial-method architecture is established [9]; the specific modal exact bound needs priority clearance. In complex-amplitude computation, XOR of secret bits is not represented by this same degree-one scalar polynomial, so there is no contradiction with ordinary one-query BV.

Canonical IDs: ALG-005, ALG-006, ALG-007.

Proposition 8.4 (Kickback and the missing character basis). *In $\mathcal{A}$, set $z = R^2 = 1 + u^2$. For the standard target flip X and $\chi = (1, z)^T$, $X\chi = z\chi$. But*

$$H_z = \begin{pmatrix} 1 & 1 \\ 1 & z \end{pmatrix} \sim \text{diag}(1, u^2)$$

is not invertible. Its n th tensor over $\mathcal{A}$ has binary rank $4 + 2n$. Multiplicative characters of C_2^n into a characteristic-two field are all trivial; an $\mathcal{A}$ -valued character matrix reduces modulo u to all ones.

Proof. Use $z^2 = 1$. Row and column elimination give the Smith diagonal displayed. Tensoring the Smith forms yields u^{2j} with multiplicity $\binom{n}{j}$; only $j = 0, 1$ survive, with binary ranks four and two. In a characteristic-two field $a^2 = 1$ implies $(a + 1)^2 = 0$ and therefore $a = 1$. Every group element in C_2^n has order at most two, proving character triviality. Every unit of $\mathcal{A}$ reduces to one, proving singular reduction for the ring character matrix. $\square$

Scope. The bit-flip target cannot have a primitive eigenvector of eigenvalue R , because applying the flip twice would give $u^2\chi = 0$. A four-level cyclic target has eigenvector entries R^{-j} and can kick back R , but its oracle is a distinct interface. Compute-control-uncompute generally costs two ordinary bit queries. Literal independent registers do not move an $\mathcal{A}$ scalar between factors automatically. The no-go is not about all finite-field transforms or odd-order groups.

Provenance and status. S-CAP Q5. Standard modular-character obstruction plus model-specific rank/algorithm comparison.

Canonical IDs: ALG-008, ALG-009, ALG-010, ALG-011, ALG-012.

Finite and supporting algorithm records. The Simon record exhausts 65,535 nonzero preparations in dimension 16 for a specified one-query, no-ancilla, $n = 2$ contract with 36 standard oracles and three shifts. It is not an arbitrary-size/adaptive Simon theorem. The conventional Grover sign and diffusion construction collapses in characteristic two, and support alone supplies no probability to amplify; this is not an all-search impossibility proof. The known Willcock–Sabry UNIQUE-SAT protocol [6] is a one-query positive control on the none-or-one-marked promise with $O(n)$ other gates; 132 promised instances for $n = 1, \dots, 6$ were reproduced. Nonlinear Boolean f still embeds as a reversible basis permutation $|x, y\rangle \mapsto |x, y + f(x)\rangle$; linear amplitude evolution does not force f to be linear. Compute-copy-uncompute is standard reversible programming [22, 7]. Printing a full state description or support table is not an allowed cost-free operational readout in QXOR.

9 The enriched LM interface

Canonical IDs: LMB-001, LMB-002.

Proposition 9.1 (Two obstructions to a bare identification). *The fixed polarity-orbit LM family is not closed under ordinary Boolean matrix composition. A unital Boolean-ring homomorphism to a local ring has image contained in $\{0, 1\}$ and cannot produce nontrivial nilpotent amplitudes.*

Proof. Take $M = |X\rangle\langle Y|$. Then

$$M^2 = \begin{pmatrix} XY & 0 \\ 0 & \neg X \neg Y \end{pmatrix},$$

which is not the polarity orbit of its upper-left entry. The finite checker finds 144 of 256 products outside the fixed family. For the second statement, the image of every Boolean element is an idempotent. In a local ring, if $e^2 = e$, one of $e, 1 - e$ is a unit; $e(1 - e) = 0$ then forces $e = 1$ or $e = 0$. $\square$

Provenance and status. S-LMB, independently checked here. Local-ring idempotents are standard; the conclusion rules out a particular canonical identification, not useful coefficient encodings.

Canonical IDs: LMB-003, LMB-004, LMB-005, LMB-007, LMB-008.

Theorem 9.2 (Sufficient symbolic support bridge). *After choosing the $\mathbb{F}_2$ -algebra $\mathcal{A}$, let $\mathcal{B}_{\mathcal{A}} = \mathcal{B} \otimes_{\mathbb{F}_2} \mathcal{A} \cong \mathcal{B}[u]/(u^4)$. Every valuation $v : \mathcal{B} \rightarrow \mathbb{F}_2$ extends as*

$$v_{\mathcal{A}} \left(\sum_{j=0}^3 f_j u^j \right) = \sum_{j=0}^3 v(f_j) u^j.$$

It commutes with enriched matrix and tensor contractions. For $q = \sum_j f_j u^j$ put $\text{Supp}(q) = \bigvee_j f_j$. Then for any chosen compatible enriched effect e and state ψ ,

$$v(\text{Supp}(e\psi)) = 1 \iff v_{\mathcal{A}}(e)v_{\mathcal{A}}(\psi) \neq 0.$$

Proof. The $\mathbb{F}_2$ basis $1, u, u^2, u^3$ gives unique enriched coefficients. Base change extends v uniquely as an $\mathcal{A}$ -algebra map and preserves finite products and sums, hence contractions. A valued $\mathcal{A}$ coefficient is nonzero exactly when at least one of its four binary coefficients is one. A Boolean valuation preserves OR, giving the equivalence. $\square$

Scope. The theorem is conditional on choosing the scalar algebra and effects; “nonzero means possible” is an added interpretation. Bare valued logical bras yield only the two one-hot rows, not the complete $\mathcal{A}$ effect family. The nonzero map is not additive ($a + a = 0$) and not multiplicative in the presence of zero divisors. For a finite valuation set Ω , this enrichment is the extensional power $\mathcal{A}^{\Omega}$, not an efficient compression theorem. General mixed-characteristic K need not admit this $\mathbb{F}_2$ tensor construction.

Provenance and status. S-LMB; S-CAP F8; ordinary base change and homomorphism preservation. Useful semantic interface, not a derivation of all measurement postulates.

Canonical IDs: LMB-006, NEG-002.

Proposition 9.3 (Division-free basis branches). *For an invertible coordinate readout B_0 over $\mathcal{B}_{\mathcal{A}}$ and coordinate projectors P_i , define $J_i = B_0^{-1} P_i B_0$. Then*

$$J_i^2 = J_i, \quad J_i J_j = 0 \ (i \neq j), \quad \sum_i J_i = I.$$

The branch $J_i \psi$ is nonzero iff the i th readout coordinate is nonzero; row-unit rescaling leaves J_i unchanged. Valuation commutes with the branch formula whenever B_0 is symbolically invertible.

Proof. Conjugate the coordinate-projector identities. A column of B_0^{-1} is unimodular, since some row of B_0 pairs with it to one. Multiplying that column by a nonzero scalar cannot give zero, proving branch/nonzero equivalence. Diagonal row-unit scaling commutes with P_i and cancels in the conjugation. Valuation preserves the inverse identity and therefore the full expression. $\square$

Scope. No division by an outcome amplitude is used. No probability, normalization or closure of primitive preparations under conditioning follows. A category of instruments or effectus requires additional axioms and proofs.

10 Further supporting no-go and finite evidence records

Canonical IDs: RES-010, RES-011, RES-012.

Proposition 10.1 (Cloning, qualified deletion and exact syndromes). *For $V = \mathbb{F}_2^d$, $d \geq 2$, no linear map clones every nonzero vector as $v \mapsto v \otimes v$. No injective deterministic linear map deletes a copy as $v \otimes v \mapsto v \otimes b$ with fixed nonzero blank b and no input-dependent environment. For an injective code map E and errors F_a , an exact decoder retaining distinct syndromes exists on their joint image iff each $F_a E$ is injective and their images form a direct sum.*

Proof. Cloning $e_i + e_j$ requires cross terms absent from the linear sum of the two clones. Duplicated vectors span the subspace generated by $e_i \otimes e_i$ and $e_i \otimes e_j + e_j \otimes e_i$, of dimension $d(d+1)/2 > d$; their proposed deleted outputs span at most d dimensions, contradicting injectivity. For syndromes, distinct retained labels occupy independent subspaces and must preserve each input, proving necessity. Conversely use the inverse of each $F_a E$ on its own independent image and append its label; these maps combine to an isomorphism on the direct sum. $\square$

Scope. Linearity alone does not forbid deletion: the noninjective map $D(e_i \otimes e_j) = \delta_{ij} e_i \otimes b$ deletes all duplicated $\mathbb{F}_2$ vectors. It is not automatically a complete deterministic allowed process. The syndrome criterion is not the general degenerate quantum error-correction criterion. Remote preparation and swapping are the standard contraction identities in M^T and MEN in compatible conventions; over rings, primitive preimage conditions still matter.

Provenance and status. S-CAP F3–F6. Generic modal no-go results are prior art [3, 10]; syndrome distinctions are supporting algebra, not a fault-tolerance threshold claim.

Canonical IDs: MOD-008, RES-013, NEG-001, NEG-003, NEG-004.

Finite GHZ and probability records. The tested three-setting GHZ support has 27 contexts, 112 possible events and no compatible assignment among 512 assignments. Within that family, 101,583 subsets of at most five contexts were checked without contradiction; a six-context witness is $ZZZ, ZXX, XZY, XXX, YYZ, YYY$. This is a finite diagnostic, not a theorem that every GHZ scenario requires six contexts. The restricted orthogonal-basis version has global assignments.

For the specified nine-context Bell support, 24 events are possible, but normalized nonnegative real no-signalling constraints force six of them to zero. The supplied exact affine constraints are the certificate; the LP is a check. A weak completion with 18 positive events and a CHSH value of four is not a faithful probability model of all original events, still less a Born rule. Schumacher–Westmoreland already discuss weak versus strong probability resolutions [4]; this is retained as background validation, not new foundational priority. Historical signed C_8/C_{16} claims are not silently imported into the Boolean core.

11 Compiler evidence is not an algebraic performance theorem

Canonical IDs: CMP-001–006, NEG-005, NEG-006.

The supplied raw file has 10,500 rows, with 10,389 fully admitted four-arm comparisons and 111 other status rows. Every fully admitted case ties CM and generic two-support optimization on final degree, final monomials, multiplications, output unique nodes, pair terms, peak monomials and serialized bytes. The independent recount found no differences. S1’s SC-A symbolic-work mechanism gate passes; the corrected S2 construction verifies requested support rather than retaining the superseded generator’s 55.161 percent rate.

These observations show that avoiding early ANF expansion can save work, but not that the final simplification is uniquely accessible to CM. Equal reductions do not imply equal recognition overhead. The source package does not supply a completed natural-workload SC-B/C/D campaign, a confirmatory SC-E abstention result, or Paper B’s separate frozen Gate P evaluation. No fresh timing campaign was run in this consolidation. The 12 benchmark tests are distinct from Paper B’s reported focused 105-test implementation suite, which was not rerun here.

12 Reproducibility and remaining proof/priority boundaries

An independent standard-library checker authored for this consolidation reconstructs all 65,536 $\mathcal{A}$ resource ranks and Smith classes, classifies all 255 nonzero dual-number resources by explicit support plus SCC 2-SAT, checks 14 actual dense decoders and enumerates all 256 fixed-LM products. The dual-number counts are 81 local, 72 logical-not-strong and 102 strong, using 62,463 SAT calls. This is a different-method finite check of the generalized theorem.

Fresh named pytest totals are 31 audit, 59 across four historical Boolean suites, and 12 benchmark tests, hence 102. Other source scripts perform exact scans and are recorded separately. An interrupted aggregate driver is retained as such; individual successful stages, not the interrupted driver, support the reported results. Original input hashes, stage logs and explicit source locations are provided in the companion repository. Presence of a file copied from a rerun working directory alone does not certify that its producing stage was rerun.

No novel theorem is claimed merely because targeted web searches found no exact match. The finite-chain-ring classification and codebook synthesis, two-setting obstruction and exact characteristic-two query bounds remain the principal candidates for specialist priority review. Drafts can be built from these proofs now, but submission-strength priority and compiler performance language must respect the remaining gaps.

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